Formulas Conversion

DOCX-export test formulas (MathJax syntax) for the LatexUtils + pandoc-service math fixes.

Plain-TeX matrix macro + \cr, multi-line (\pmatrix -> pmatrix env, \cr -> \\,
 handling):

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Plain-TeX \matrix macro (\matrix -> matrix env, no delimiters):

$$I = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

Plain-TeX \cases macro (\cases -> cases env):

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

\eqalign (-> aligned env, relation alignment on &):

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \\ (a-b)^2 &= a^2 - 2ab + b^2 \end{aligned}$$

\displaylines (-> gathered env, centred lines):

$$\begin{gathered} x + y = 10 \\ x - y = 2 \end{gathered}$$

Infix \over (-> \frac):

$$\frac{a+b}{c+d}$$

Infix \atop (-> \substack, ruleless stack):

$$\begin{matrix} n \\ k \end{matrix}$$

\overwithdelims (-> \left D1 \frac \right D2, delimited ruled fraction):

$$\left[\frac{a}{b}\right]$$

\atopwithdelims (-> \left D1 \substack \right D2, the binomial coefficient):

$$\binom{n}{k}$$

\cfrac continued fraction (-> \frac, incl. nested):

$$\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{x}}}$$

\root ... \of ... (-> \sqrt[index]{radicand}, grouped index and radicand):

$$\sqrt[n+1]{x+y}$$

\buildrel ... \over ... (-> \overset), also exercises the \rm font switch in the top:

$$A \overset{\text{def}}{=} B$$

\require{...} directive stripped, so the native \cancel survives:

$$\frac{\cancel{3}\cdot 3}{\cancel{3}} = 3$$

Old font switches \rm / \bf / \it / \sf / \tt (-> \mathrm / \mathbf / \mathit / \mathsf / \mathtt):

$$\mathrm{d}x + \mathbf{v}\cdot t + \mathsf{S} + \mathtt{code}$$

Old font switches \cal / \frak / \Bbb / \scr (-> \mathcal / \mathfrak / \mathbb / \mathscr):

$$\mathcal{L} \in \mathfrak{g}, \quad x \in \mathbb{R}, \quad \mathscr{F}(x)$$

Math color \color{NAME}{X} (named colors, pandoc-service shim):

$$\color{red}{E} = \color{blue}{mc^2}$$

Math color \color{NAME}{X} on a larger sub-expression:

$$\int_0^1 \color{green}{x^2}\ dx = \color{orange}{\frac{1}{3}}$$

Math color by hex "#RRGGBB" (an HTML color string, accepted by MathJax v2 \color):

$$\color{#ffa500}{\alpha} + \color{#3cb371}{\beta}$$

Nested math color (innermost color wins per segment):

$$\color{red}{a + \color{blue}{b + \color{green}{c} + d}}$$

ADVANCED: quadratic formula with a colored discriminant (\color inside \sqrt inside \frac):

$$x = \frac{-b \pm \sqrt{\color{red}{b^2 - 4ac}}}{2a}$$

ADVANCED: matrix macro with \over inside cells, multi-line (matrix + \cr + \over-in-cell +
):

$$\begin{pmatrix} \frac{a}{b} & c \\ d & \frac{e}{f} \end{pmatrix}$$

ADVANCED: real-world \atop as a two-line subscript under a sum (-> \substack):

$$\sum_{\substack{0\le i\le m\\ 0<j<n}} P\left(i,j\right)$$

ADVANCED: \eqalign combined with color and \over, multi-line:

$$\begin{aligned} E &= \frac{mc^2}{1} \\ p &= mv \end{aligned}$$

ADVANCED: font switch + color + \root + \frac in one expression:

$$\boldsymbol{F} = \mathrm{m} \cdot \sqrt[3]{\frac{a}{b}}$$

NATIVE sanity (texmath handles these already; confirm they still render): \choose:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

NATIVE sanity: decorations \underbrace / \xrightarrow / \overset:

$$a + a + \cdots + a = na \xrightarrow{f} \overset{!}{=} b$$

$$\underbrace{}_{n \text{ times}}$$

FAILED TO CONVERT \ce (mhchem):

$$\ce{C2H5OH}$$

FAILED TO CONVERT \def:

$$\def\half{\frac{1}{2}}\half$$